

GEILO-OLDEBRATEN, Norway

WMO: 013590

Lat: 60.530N

Lon: 8.195E

Elev: 2536

StdP: 13.40

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.2	(c) -6.0	(d) -16.2	(e) 2.5	(f) -10.5	(g) -11.1	(h) 3.3	(i) -5.6	(j) 25.4	(k) 32.2	(l) 21.9	(m) 33.1	(n) 1.3	(o) 220	(p) 0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	73.0	56.1	69.1	55.2	65.2	53.2	58.6	68.4	56.8	65.3	55.1	62.9	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.0	70.8	60.7	53.1	65.8	59.2	51.2	61.4	57.4	26.5	68.5	25.3	65.1	24.2	62.9	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.1	16.8	14.3	DB	-17.1	77.4	6.4	3.9	-21.7	80.2	-25.5	82.4	-29.1	84.6	-33.8	87.4	
			WB	-17.5	60.6	6.4	2.2	-22.1	62.1	-25.8	63.4	-29.4	64.6	-34.0	66.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	18.7	20.5	26.1	33.0	42.1	49.7	54.4	51.6	45.6	36.1	27.1	21.9	
	DBStd	14.76	11.64	11.06	9.29	6.69	6.75	5.03	5.32	4.21	5.10	6.42	9.06	11.94	
	HDD50	5559	970	825	741	509	262	63	16	30	149	434	688	872	
	HDD65	10714	1435	1245	1206	959	710	458	330	416	583	897	1138	1337	
	CDD50	321	0	0	0	0	17	55	152	79	17	1	0	0	
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH74	62	0	0	0	0	1	5	56	0	0	0	0	0	
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0	
Wind	WSAvg	5.7	5.7	5.7	7.0	6.2	6.4	5.6	4.8	4.5	5.8	5.5	5.1	6.0	
Precipitation	PrecAvg	30.6	2.8	1.9	1.9	1.7	1.9	2.6	3.3	3.4	2.8	3.0	2.9	2.5	
	PrecMax	42.1	6.1	4.0	5.3	3.8	4.9	6.9	7.5	6.2	5.4	6.6	6.1	5.3	
	PrecMin	21.0	0.4	0.2	0.3	0.5	0.2	0.4	0.7	1.1	1.3	0.9	0.7	0.4	
	PrecStd	4.3	1.3	0.9	0.9	0.9	1.1	1.5	1.6	1.5	0.9	1.4	1.2	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	44.0	49.9	55.5	71.2	74.4	79.5	72.0	66.4	58.0	46.8	43.1	
		MCWB	37.5	38.2	41.6	44.3	53.4	54.6	59.4	56.7	53.6	50.7	41.6	39.7	
	2%	DB	39.3	40.5	43.8	50.3	64.5	70.3	75.1	67.5	62.0	52.1	43.3	41.0	
		MCWB	35.6	35.9	37.5	41.2	51.0	53.8	57.9	55.1	51.6	45.0	39.6	37.8	
	5%	DB	36.8	38.1	40.7	46.6	59.6	65.8	71.3	64.2	58.2	48.7	40.6	38.9	
		MCWB	33.5	34.2	35.8	39.3	48.5	51.0	56.8	53.9	50.1	43.4	37.2	35.6	
	10%	DB	33.6	35.3	38.1	43.6	54.7	61.4	66.7	60.8	54.5	45.8	38.5	36.6	
		MCWB	30.6	31.4	33.5	37.7	45.5	49.4	54.9	52.7	48.4	41.4	35.3	33.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	38.9	41.5	45.4	55.9	56.8	62.1	59.2	56.3	51.9	42.6	40.1
			MCDB	41.6	42.9	48.1	53.1	66.1	66.3	73.1	66.6	62.9	58.0	46.1	42.5
2%		WB	35.9	36.5	38.1	42.2	52.1	55.0	59.6	57.2	53.0	46.5	39.6	37.8	
		MCDB	39.1	40.0	42.9	48.9	62.1	66.1	70.3	63.8	59.1	50.9	42.5	40.6	
5%		WB	33.5	34.4	36.2	40.1	49.4	53.2	58.1	55.5	51.0	44.0	37.6	35.9	
		MCDB	36.7	38.0	40.5	46.0	57.8	63.0	68.3	62.1	56.4	47.8	40.3	38.8	
10%		WB	30.7	31.4	33.6	38.1	46.7	51.3	56.3	53.7	49.1	41.5	35.5	33.4	
		MCDB	33.4	35.1	37.7	43.0	54.3	59.3	65.0	59.5	53.8	45.3	38.3	36.4	
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.6	14.9	15.2	16.0	17.9	18.0	16.7	14.9	12.7	11.3	12.5	
		MCDBR	13.0	12.6	14.3	20.3	24.4	27.6	27.9	23.2	22.3	18.5	12.5	11.3	
		MCWBR	11.2	10.1	10.2	12.7	13.3	13.3	12.4	12.0	13.3	12.4	10.0	10.1	
	5% WB	MCDBR	13.0	12.1	12.9	18.1	22.3	21.7	23.6	19.4	18.6	14.8	11.8	11.5	
		MCWBR	11.3	10.1	9.6	11.6	12.8	11.8	11.2	11.2	11.8	10.8	9.6	10.3	
Clear-Sky Solar Irradiance	taub	0.210	0.250	0.283	0.317	0.324	0.321	0.335	0.327	0.297	0.279	0.222	0.180		
	taud	2.156	2.269	2.242	2.240	2.363	2.471	2.455	2.464	2.539	2.458	2.201	2.023		
	Ebn at Noon	187	244	269	279	286	289	281	275	267	232	177	144		
	Edh at Noon	12	20	29	36	34	31	31	28	22	17	11	8		
All-Sky Solar Radiation	RadAvg	95	291	745	1288	1518	1601	1465	1151	750	350	121	58		
	RadStd	19	69	83	186	197	164	183	102	94	60	25	6		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page